

| A-Team Chronicles

## A-Team Chronicles

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Networking

# OCI basic monitor/alarms setup for FastConnect and VPN using native services.

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## Introduction

Many customers asked how they can set up monitoring and send alarms for FastConnect and/or VPN tunnels if one of their connections goes down or has unexpected issues.

OCI has different out of the box services that can help accomplish this:

1. Monitoring service (Official documentation <https://docs.oracle.com/en-us/iaas/Content/Monitoring/home.htm>) that has the following components:
  - Metrics – Will show the raw data coming from different sources. For FastConnect and VPN, these sources are automatically posted by Oracle Cloud Infrastructure resources.
  - Alarms – Is triggered based on the value of the metric.
2. Notification's service (Official documentation <https://docs.oracle.com/en-us/iaas/Content/Notification/home.htm>) – Broadcasts messages to distributed components through a publish-subscribe pattern.

Now, this being said, let's dive more into these components:

VPN and FastConnect metrics official documentation can be found on [https://docs.oracle.com/en-us/iaas/Content/Network/Reference/ipsecmetrics2.htm#VPN\\_Connect\\_Metrics](https://docs.oracle.com/en-us/iaas/Content/Network/Reference/ipsecmetrics2.htm#VPN_Connect_Metrics) and <https://docs.oracle.com/en-us/iaas/Content/Network/Reference/fastconnectmetrics.htm>

Based on those documents we have the following metrics:

**VPN: TunnelState, PacketsReceived, BytesReceived, PacketsSent, BytesSent and PacketsError**

| Metric           | Metric Display Name | Unit            | Description                                                                                                                                                                                                                                    | Dimensions                       |
|------------------|---------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Tunnel State     | IPSec Tunnel State  | Binary (1 or 0) | Whether the tunnel is up (1) or down (0).                                                                                                                                                                                                      | parentResourceId<br><br>publicIp |
| Packets Received | Packets Received    | Packets         | Number of packets received at the Oracle end of the connection.                                                                                                                                                                                |                                  |
| Bytes Received   | Bytes Received      | Bytes           | Number of bytes received at the Oracle end of the connection.                                                                                                                                                                                  |                                  |
| Packets Sent     | Packets Sent        | Packets         | Number of packets sent from the Oracle end of the connection.                                                                                                                                                                                  |                                  |
| Bytes Sent       | Bytes Sent          | Bytes           | Number of bytes sent from the Oracle end of the connection.                                                                                                                                                                                    |                                  |
| Packets Error    | Packets with Errors | Packets         | Number of packets dropped at the Oracle end of the connection. Dropped packets indicate a misconfiguration in some part of the overall system. Check if there's been a change to the configuration of your VCN, Site-to-Site VPN, or your CPE. |                                  |

**FastConnect: BitsReceived, BitsSent, BytesReceived, BytesSent, ConnectionState, PacketsError, PacketsDiscarded, PacketsReceived, PacketsSent, Ipv4BgpSessionState and Ipv6BgpSessionState**

| Metric                 | Metric Display Name     | Unit            | Description                                                                                                                                                                                            | Dimensions                                                           |
|------------------------|-------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>BitsReceived</b>    | <b>Bits Received</b>    | Bits            | <p>Number of bits received on the FastConnect interface at the Oracle end of the connection.</p> <p>For a cross-connect group (LAG), the value is the sum across all cross-connects in the group.</p>  | <b>component</b><br><br><b>resourceId</b><br><br><b>resourceName</b> |
| <b>BitsSent</b>        | <b>Bits Sent</b>        | Bits            | <p>Number of bits sent from the FastConnect interface at the Oracle end of the connection.</p> <p>For a cross-connect group (LAG), the value is the sum across all cross-connects in the group.</p>    |                                                                      |
| <b>BytesReceived</b>   | <b>Bytes Received</b>   | Bytes           | <p>Number of bytes received on the FastConnect interface at the Oracle end of the connection.</p> <p>For a cross-connect group (LAG), the value is the sum across all cross-connects in the group.</p> |                                                                      |
| <b>BytesSent</b>       | <b>Bytes Sent</b>       | Bytes           | <p>Number of bytes sent from the FastConnect interface at the Oracle end of the connection.</p> <p>For a cross-connect group (LAG), the value is the sum across all cross-connects in the group.</p>   |                                                                      |
| <b>ConnectionState</b> | <b>Connection State</b> | Binary (1 or 0) | <p>The values are up (1) or down (0).</p> <p>For a virtual circuit, the operational state of the virtual circuit's</p>                                                                                 |                                                                      |

|                          |                            |         |                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|----------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |                            |         | <p>interface.</p> <p>For a cross-connect group, this reflects the overall operational state of the cross-connects that make up the cross-connect group (LAG). If at least one of the cross-connects is up, this value is up (1). If <i>all</i> the cross-connects in the group are down, this value is down (0).</p>                                          |
| <b>Packets Error</b>     | <b>Packets with Errors</b> | Packets | <p>Number of packets dropped at the Oracle end of the connection. Dropped packets indicate a misconfiguration in some part of the overall system. Check if there's been a change to the configuration of your VCN, the virtual circuit, or your CPE.</p> <p>For a cross-connect group (LAG), the value is the sum across all cross-connects in the group.</p> |
| <b>Packets Discarded</b> | <b>Packets Discarded</b>   | Packets | <p>Number of packets discarded at the Oracle end of the connection.</p>                                                                                                                                                                                                                                                                                       |
| <b>Packets Received</b>  | <b>Packets Received</b>    | Packets | <p>Number of packets received on the FastConnect interface at the Oracle end of the connection.</p> <p>For a cross-connect group (LAG), the value is the sum across all cross-connects in the group.</p>                                                                                                                                                      |
| <b>Packets Sent</b>      | <b>Packets Sent</b>        | Packets | <p>Number of packets sent from the FastConnect interface at the Oracle end of the connection.</p> <p>For a cross-connect group (LAG), the value is the sum across all cross-connects in the group.</p>                                                                                                                                                        |

|                                      |                                   |                    |                                                                                              |
|--------------------------------------|-----------------------------------|--------------------|----------------------------------------------------------------------------------------------|
| <b>Ipv4Bgp<br/>Session<br/>State</b> | <b>IPv4 BGP<br/>Session State</b> | Binary (1<br>or 0) | The values are up (1) or down (0). The status of the IPv4 BGP session for a virtual circuit. |
| <b>Ipv6Bgp<br/>Session<br/>State</b> | <b>IPv6 BGP<br/>Session State</b> | Binary (1<br>or 0) | The values are up (1) or down (0). The status of the IPv6 BGP session for a virtual circuit. |

Before moving to the Alarm configuration, we need to understand some basics about the Metrics service. Metrics are pulled from the infrastructure at 1 min interval for VPN and 4 min for FastConnect. Based on this information, we should precisely understand the threshold value that we should put in place so we don't receive false alarms.

Based on all the metrics, we can configure alarms, and when they are triggered, a notification will be sent.

Before showing how to implement the solution, let's discuss the prerequisites and scenarios we can implement using this blog.

## Prerequisites:

OCI VCN configured and working with traffic between On-premises and OCI resources

FastConnect and/or IPsec configured and working

## Scenarios that can be implemented:

FastConnect virtual circuit down (Connection State)

Fastconnect BGP down (Session State)

FastConnect errors (Packets with errors)

FastConnect bandwidth throughput

IPSec Down (Connection state)

IPSec BGP down (only tunnels with BGP)

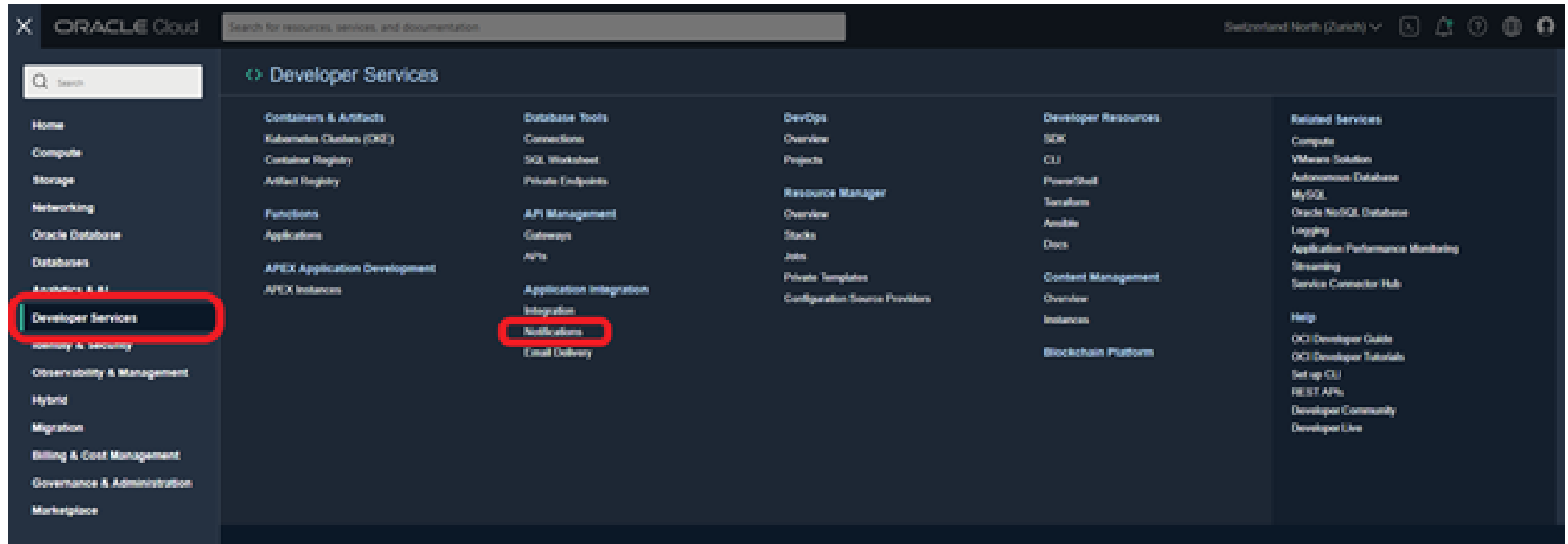
IPSec Errors

## Solution description:

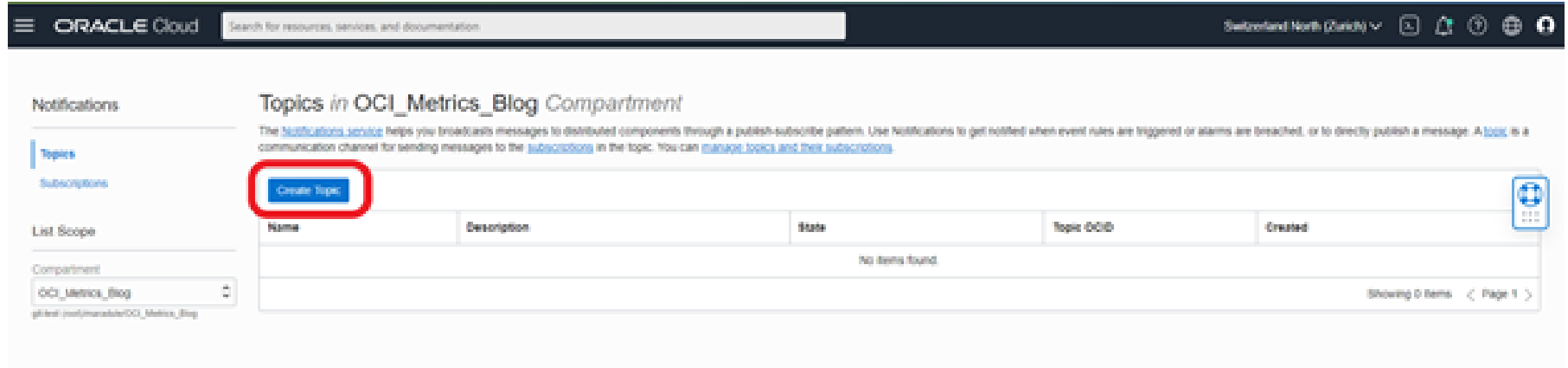
1. Setting up the Notification service. We can create one or many notification services that we can use, depending on the requirements (team to be alerted, distribution channel, etc.). In our case, we will provision and use only one Notification service. To provision such a service, we need to do the following things:

Go to Notification by navigating to **"Burger menu"** -> **"Developer Services"** -> **"Notifications"**





Click on the “**Create Topic**” button



Add a name and a description (optional) and click “**Create**”

## Create Topic [Help](#)

To create a topic in a different compartment, [click here](#).

Name  
OCI\_Metrics\_Blog\_Topic

Description Optional  
Topic used for OCI Metric Blog

[Show Advanced Options](#)

Once the topic is created, an administrator needs to create an identity policy to enable access.

[Create](#) [Cancel](#)

Now, we will see the topic we have just created in the list of topics

### Topics in OCI\_Metrics\_Blog Compartment

The [Notifications service](#) helps you broadcast messages to distributed components through a publish-subscribe pattern. Use Notifications to get notified when event rules are triggered or alarms are breached, or to directly publish a message. A [topic](#) is a communication channel for sending messages to the [subscriptions](#) in the topic. You can [manage topics and their subscriptions](#).

[Create Topic](#)

| Name                          | Description                    | State  | Topic OCID                                              | Created                        |
|-------------------------------|--------------------------------|--------|---------------------------------------------------------|--------------------------------|
| <b>OCI_Metrics_Blog_Topic</b> | Topic used for OCI Metric Blog | Active | ...dctnugawoa <a href="#">Show</a> <a href="#">Copy</a> | Wed, Mar 9, 2022, 05:32:08 UTC |

Showing 1 item < Page 1 >

Click in the topic name that we have created, which will bring us to the subscription page. On this page, click on **“Create Subscription”**

ORACLE Cloud Search for resources, services, and documentation Switzerland North (Zurich)

Notifications > Topics > Topic Details

## OCI\_Metrics\_Blog\_Topic

[Publish Message](#) [Move Resource](#) [Delete](#) [Add Tags](#)

**Topic Information** **Tags**

**Description:** Topic used for OCI Metric Blog

**OCID:** ...dctnugawoa [Show](#) [Copy](#)

**Short Topic ID:** [Show](#) [Copy](#)

**Compartment:** gd-test-prod/iammodule/OCI\_Metrics\_Blog

**Created:** Wed, Mar 9, 2022, 05:32:08 UTC

**Resources**

This list shows only resources that you have access to.

[Create Subscription](#)

| Subscription OCID | State | Protocol | Endpoint | Created |
|-------------------|-------|----------|----------|---------|
| No items found.   |       |          |          |         |

Showing 0 items < Page 1 >

Now select the protocol to use from the list. At the moment of writing this blog, we have the following possibilities: **“Email”**, **“Function”**, **“HTTPS (custom URL)”**, **“PagerDuty”**, **“Slack”** and **“SMS (for Monitoring and Service Connector Hub)”**. In our blog, I will select **“Email”** and I will add my email address and click **“Create”**

## Create Subscription [Help](#)

View [steps for creating subscriptions](#) and learn about [supported subscription protocols](#).

### Configure Subscription

Protocol

Email

Email

example@example.com

 Email notifications use the sender "noreply" at a region-specific notification domain.  
Example sender: noreply@notification.us-ashburn-1.oci.oraclecloud.com  
[Creating a subscription for Email](#)

[Show Advanced Options](#)

Create

Cancel

Once you do this, you will receive an email to confirm the subscription. Confirmation is needed so the system can activate the delivery of the emails.

## Oracle Cloud Infrastructure Notifications Service Subscription Confirmation



noreply@notification.eu-zurich-1.oci.oraclecloud.com  
To: Marius Radulescu



Reply Reply All Forward ...  
miercuri 09.03.2022 08:13

You have chosen to subscribe to the topic:

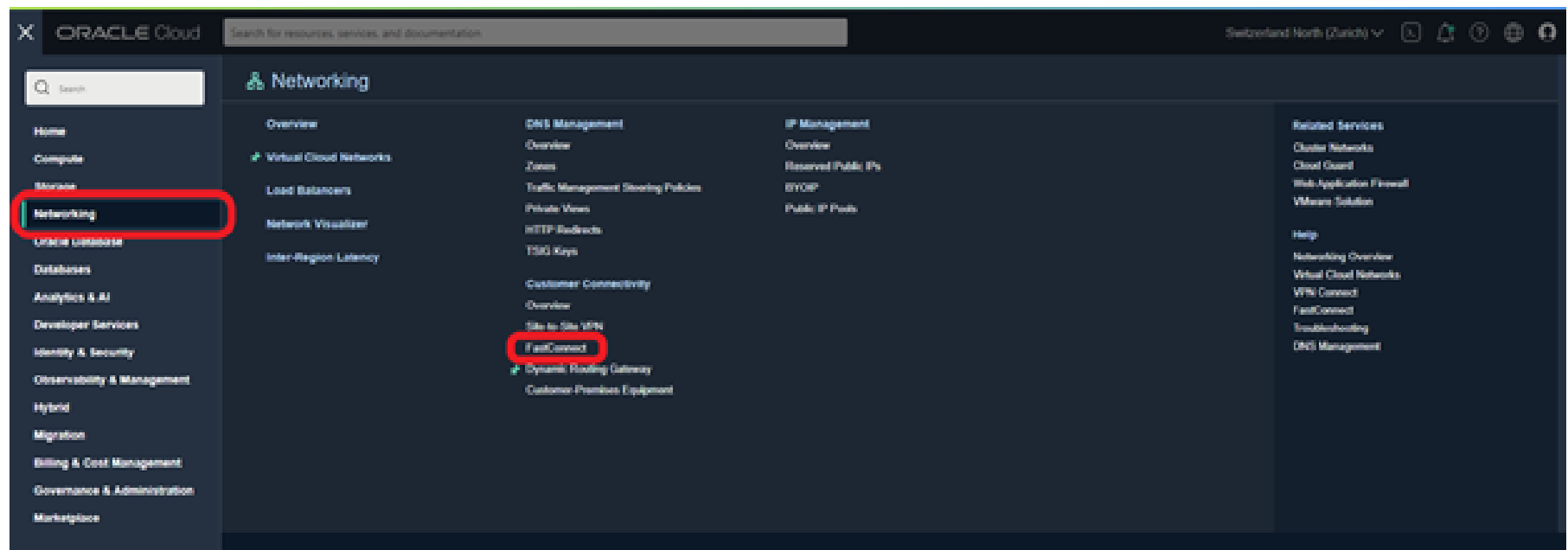
**OCI\_Metrics\_Blog\_Topic** (Topic OCID: ocid1.onstopic.oc1.eu-zurich-1.aaaaaaaqdgwdfhjoyw2lenjpcddq6nsc12e3y7sydr524studctnulgwoa)

To confirm this subscription, click or visit the link below (If this was in error, you can ignore this message):

[Confirm subscription](#)

2. Now that the Notification service is created, we can move on to how to create alarms for FastConnect by doing the following tasks:

From the burger menu, go to “**Networking**” – > “**Fastconnect**”



Now select the Fastconnect we want to activate Monitoring

### FastConnect Connections in OCI\_Metrics\_Blog Compartment

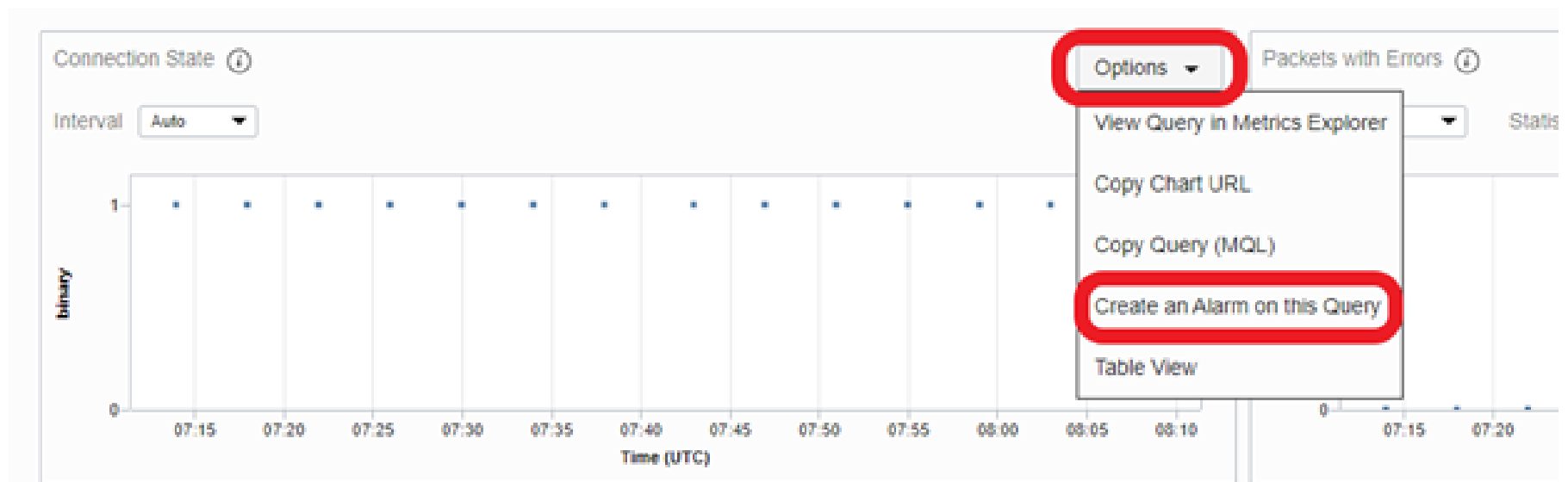
FastConnect is a connection between a customer's on-premises network and Oracle Cloud Infrastructure over a private physical network instead of the internet.

Create FastConnect

| Name                                  | Lifecycle State ⓘ | IPv4 BGP State ⓘ | IPv6 BGP State ⓘ | Connection Type ⓘ | Created                        |
|---------------------------------------|-------------------|------------------|------------------|-------------------|--------------------------------|
| OCI_Metrics_Blog_OCI_Side_FastConnect | Provisioned       | Up               | Down             | Partner           | Mon, Mar 7, 2022, 11:16:42 UTC |

Showing 1 item < 1 of 1 >

Now, we can see Metrics for the FastConnect. On this, we can choose the metric we want and create the alarm, for example, on the **“Connection State”**, by going on **“Options”** –> **“Create an Alarm on this Query”** as shown in this example:



This action will take us on the **“Create Alarm”** page that will need to be filled up with the following information:

- Alarm Name
- Alarm Severity
- Alarm Body (optional)

- d. Tags (optional)
- e. Metric description (can remain untouched)
- f. Metric dimensions (can remain untouched)
- g. Trigger rule – in our case for “Connection State” we will choose “less than” operator with a “Value” of 1 and a “Trigger delay minutes” of 5
- h. ConnectionState Graph (untouched)
- i. Notifications – we select the topic created at point 1, in our case, “OCI\_Metrics\_Blog\_Topic” Also, in this section, we can select to repeat notification in case the alarm remains firing more than a specified interval. In our case, I have selected 60 minutes. Here we can also select a suppression on this alarm to send notification from “Suppress Notification” and select a start date and an end date of this suppression. “Repeat notification” and “Suppress notifications” are optional fields.
- j. Enable Alarm
- k. Save Alarm

### Define alarm

Switch to Advanced Mode

Alarm name

OCI\_Metrics\_FC\_ConnState\_Alarm

Alarm severity ⓘ

Critical

Alarm body ⓘ ⓘ

Enter notification content. Example: High CPU usage alert. Follow runbook instructions for resolution.

Limited to 1000 characters (0/1000)

#### Tags (optional)

Tagging is a metadata system that allows you to organize and track alarms within your tenancy.

| Tag namespace               | Tag key         | Value                 |
|-----------------------------|-----------------|-----------------------|
| None (apply a freeform tag) | Enter a tag key | Enter a tag key first |

+ Additional tag

#### Metric description

The metric to evaluate for the alarm.

|                                                                                                         |                                                          |                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <div>Compartment</div> <div>OCI_Metrics_Blog</div> <div>glt-test (root)/maradule/OCI_Metrics_Blog</div> | <div>Metric namespace ⓘ</div> <div>oci_fastconnect</div> | <div>Resource group ⓘ ⓘ</div> <div>No resource group</div> <div>No resource groups for selected compartment and namespace</div> |
| <div>Metric name ⓘ</div> <div>ConnectionState</div>                                                     | <div>Interval ⓘ</div> <div>1m</div>                      | <div>Statistic ⓘ</div> <div>Mean</div>                                                                                          |



### Metric dimensions

Dimension name ⓘ  
resourceId

Dimension value ⓘ  
ocid1.virtualcircuit.oc1.eu-zurich-1.aaaaaaaajucqarcygsgbxwoy3vmcszunfwtaibn3yptam2xchthem7h3q

☐ Aggregate metric streams

+ Additional dimension

### Trigger rule

The condition for putting the alarm in the firing state.

operator ⓘ  
less than

Value ⓘ  
1

Trigger delay minutes ⓘ  
1

ConnectionState

binary

Time (UTC)

Show Data Table

Query 1 (1 metric streams)

ConnectionState[resourceId = "ocid1.virtualcircuit.oc1.eu-zurich-1.aaaaaaaajucqarcygsgbxwoy3vmcszunfwtaibn3yptam2xchthem7h3q"] as conn

### Notifications

#### Destinations

Destination service ⓘ  
Notifications Service

Compartment ⓘ  
OCI\_Metrics\_Blog  
git-test (root)/maradule/OCI\_Metrics\_Blog

Topic ⓘ  
OCI\_Metrics\_Blog\_Topic  
Create a topic

Message Format ⓘ  
☒ Send formatted messages  
☐ Send Pretty JSON messages (raw text with line breaks)  
☐ Send raw messages

+ Additional destination service

☒ Repeat notification? (if alarm continues to fire) ⓘ  
Notification frequency  
60 minutes

☒ Suppress notifications  
Suppression description Optional Start time End time  

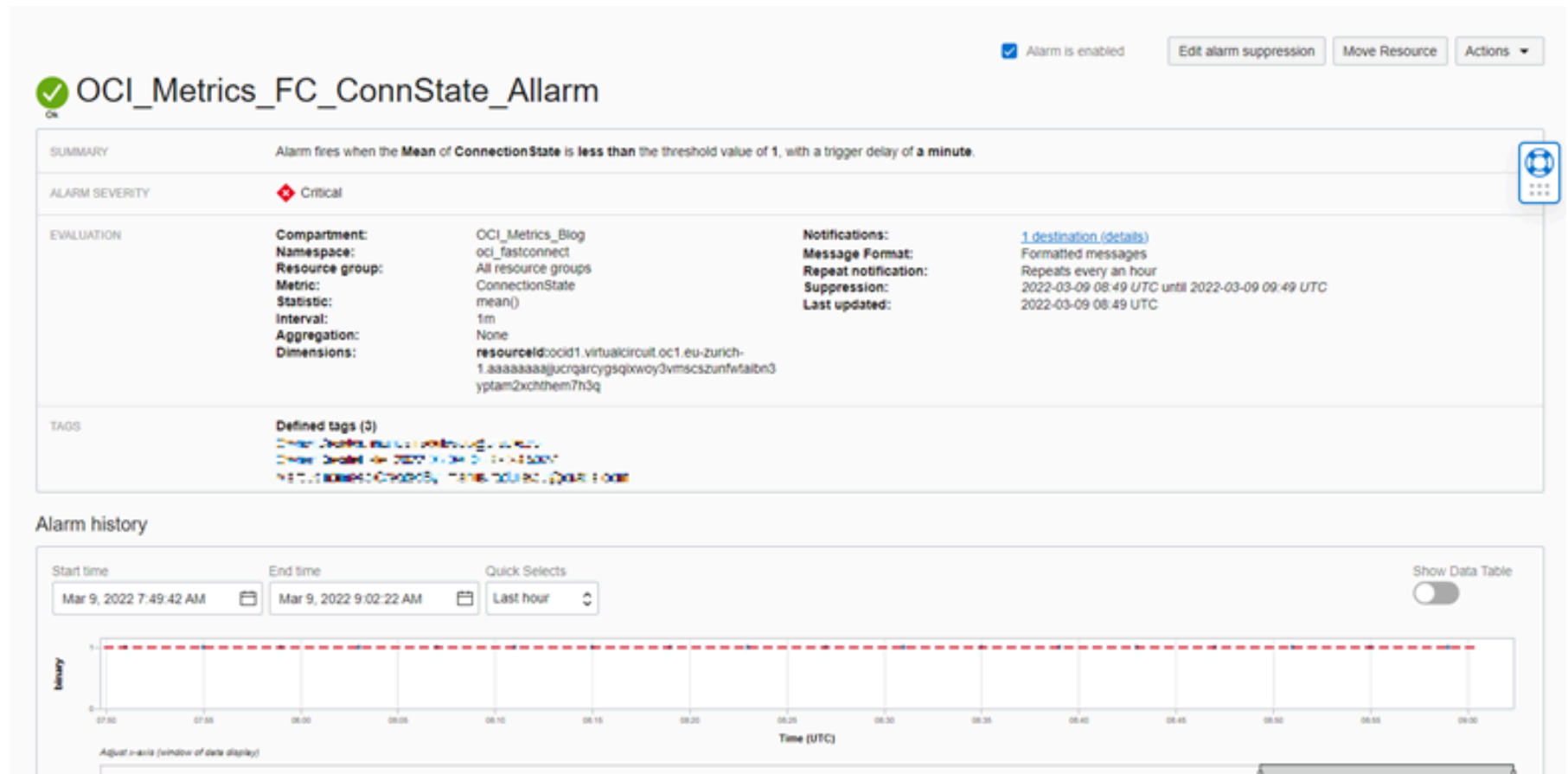
Mar 9, 2022 8:33:19 AM Mar 9, 2022 9:33:19 AM

☒ Enable this alarm? ⓘ  
This alarm applies to 1 metric streams.

Save alarm

Cancel

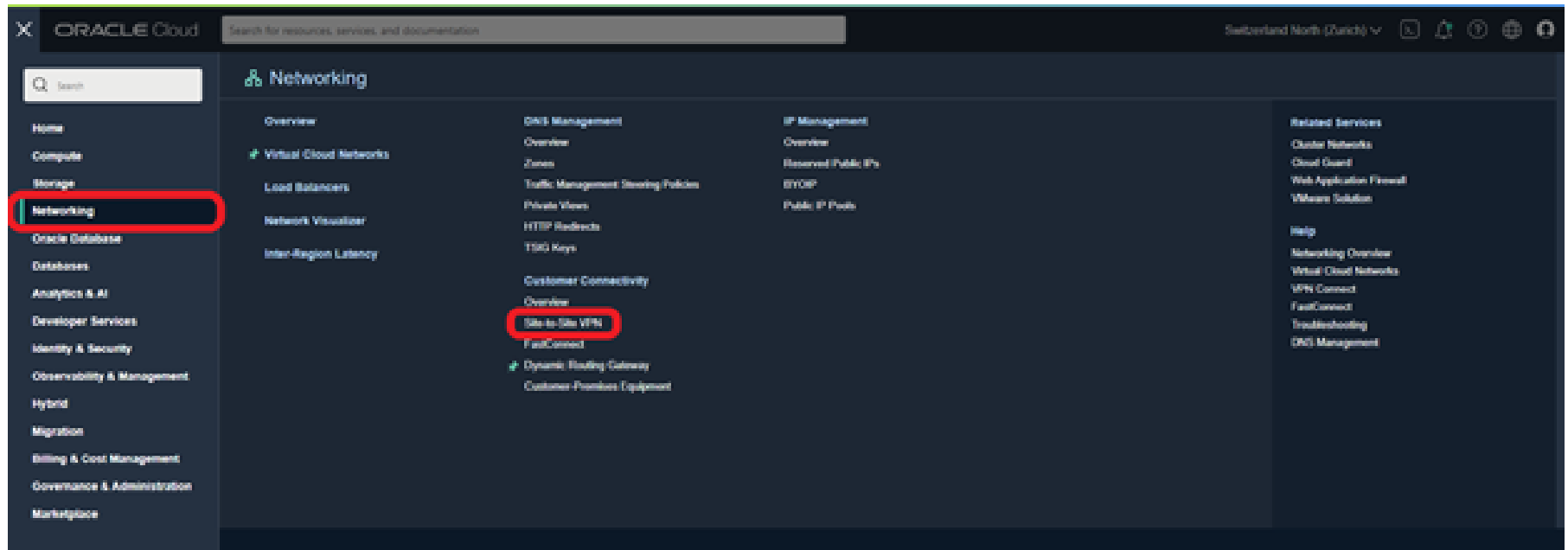
After creation Alarm page should look like this



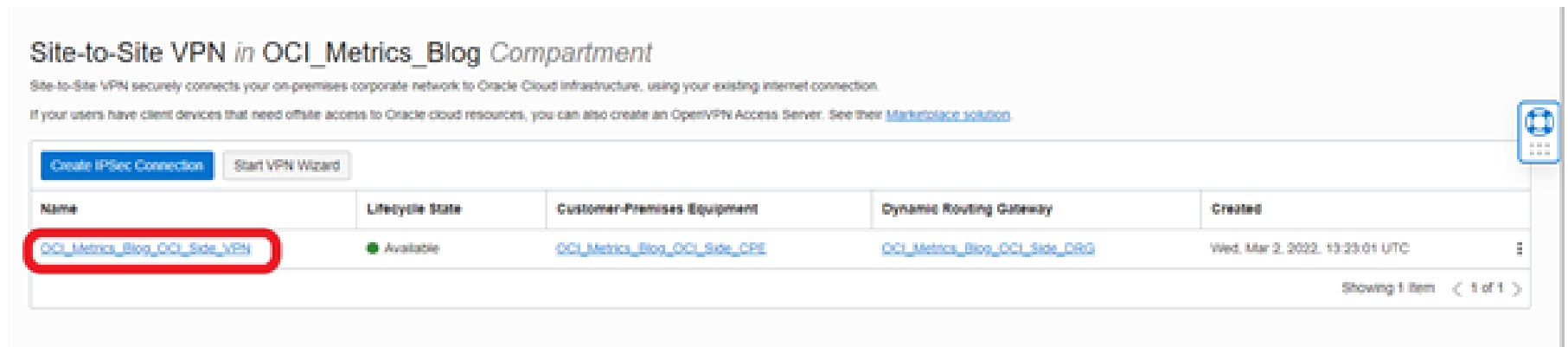
Based on this template, we can create multiple alarms that have a different starting point from any of the metrics available on the FastConnect ( BitsReceived, BitsSent, BytesReceived, BytesSent, ConnectionState, PacketsError, PacketsDiscarded, PacketsReceived, PacketsSent, Ipv4BgpSessionState and Ipv6BgpSessionState)

3. Now, we can also make an alarm for one of the VPN tunnels. For doing this, we will need to do the following tasks:

From the burger menu, go to **“Networking”** – > **“Site-to-Site VPN”**



Select the VPN we want to create an Alarm for



Select the VPN tunnel. In our case, I will select the tunnel with BGP so I can show how to create an Alarm for the BGP session

### OCI\_Metrics\_Blog\_OCI\_Side\_VPN

[Edit](#)
[Choose New Compartment](#)
[Add Tags](#)
[Open CPE Configuration Helper](#)
[Terminator](#)

[IPSec Connection Information](#)
[CPE & Tunnels Information](#)
[Tags](#)

**Static Route CIDR:** 192.168.0.0/24 ⓘ  
**Created:** Wed, Mar 2, 2022, 13:23:01 UTC  
**VPN Connect Version:** v2 ⓘ

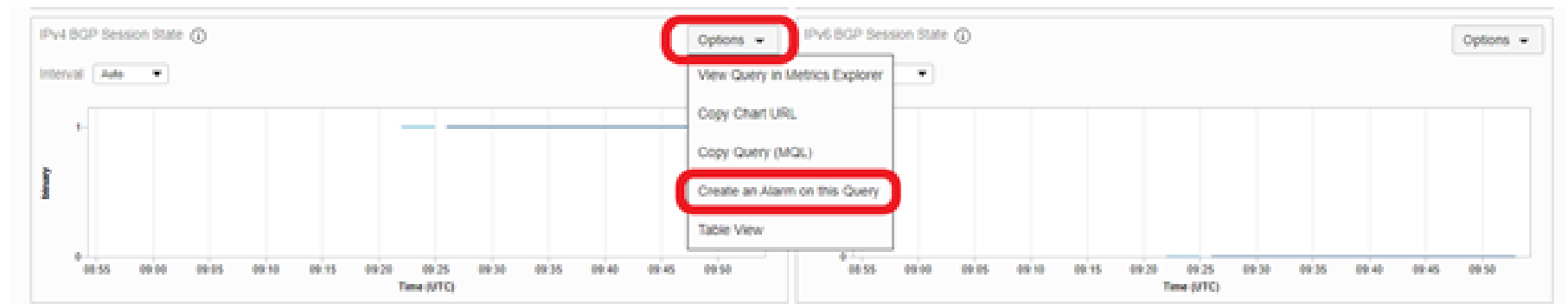
**OCID:** ...ofbca [Show](#) [Copy](#)  
**BRG:** ...mlDea [Show](#) [Copy](#) [View](#)  
**CPE:** ...o7xna [Show](#) [Copy](#) [View](#)

### Tunnels in OCI\_Metrics\_Blog Compartment

| Name                             | Lifecycle State ⓘ | IPSec Status ⓘ | IPv4 BGP Status ⓘ | IPv6 BGP Status ⓘ | Oracle VPN IP Address | Routing Type        |
|----------------------------------|-------------------|----------------|-------------------|-------------------|-----------------------|---------------------|
| <a href="#">Tunnel1 - BGP</a>    | Available         | Up             | Up                | Down              | 140.236.169.191       | BGP Dynamic Routing |
| <a href="#">Tunnel2 - Static</a> | Available         | Up             | -                 | -                 | 152.67.68.187         | Static Routing      |

Showing 2 items

Scroll down to the IPv4 BGP Session State click on **“Options”** -> **“Create an Alarm on this Metric”**



This action will take us on the **“Create Alarm”** page that will need to be filled up with the following information:

- Alarm Name
- Alarm Severity
- Alarm Body (optional)

- d. Tags (optional)
- e. Metric description (can remain untouched)
- f. Metric dimensions (can remain untouched)
- g. Trigger rule – in our case for “Connection State” we will choose “less than” operator with a “Value” of 1 and a “Trigger delay minutes” of 5
- h. ConnectionState Graph (untouched)
- i. Notifications – we select the topic created at point 1, in our case, “OCI\_Metrics\_Blog\_Topic” Also, in this section, we can select to repeat notification in case the alarm remains firing more than a specified interval. In our case, I have selected 60 minutes. Here we can also select a suppression on this alarm to send notification from “Suppress Notification” and select a start date and an end date of this suppression. “Repeat notification” and “Suppress notifications” are optional fields.
- j. Enable Alarm
- k. Save Alarm

### Define alarm

Switch to Advanced Mode

Alarm name

OCI\_Metrics\_VPN\_Tunnel1\_BGPSessionState\_Alarm

Alarm severity ⓘ

Critical

Alarm body ⓘ Optional ⓘ

Enter notification content. Example: High CPU usage alert. Follow runbook instructions for resolution.

Limited to 1000 characters (0/1000)

Tags (optional)

Tagging is a metadata system that allows you to organize and track alarms within your tenancy.

Tag namespace

None (apply a freeform tag)

Tag key

Enter a tag key

Value

Enter a tag key first

+ Additional tag

Metric description

The metric to evaluate for the alarm.

Compartment

OCI\_Metrics\_Blog

glt-test (root)/maradule/OCI\_Metrics\_Blog

Metric namespace ⓘ

oci\_vpn

Resource group ⓘ Optional ⓘ

No resource group

No resource groups for selected compartment and namespace

Metric name ⓘ

Ipv4BgpSessionState

Interval ⓘ

1m

Statistic ⓘ

Mean

### Metric dimensions

Dimension name ⓘ  
publicip

Dimension value ⓘ  
140.238.169.191

Dimension name ⓘ  
parentResourceid

Dimension value ⓘ  
ocid1\_ipsecconnection.oc1.eu-zurich-1.aaaaaaaalvchpf4vwdngofstjo3xrqnstegj4uv2o5tyw6g2vn5of3oca

☐ Aggregate metric streams

+ Additional dimension

### Trigger rule

The condition for putting the alarm in the firing state.

operator ⓘ  
less than

Value ⓘ  
1

Trigger delay minutes ⓘ  
1

Ipv4BgpSessionState

binary



Time (UTC)

Show Data Table

☐

Query 1 (2 metric streams)

Ipv4BgpSessionState[?publicIp="140.238.169.191",parentResourceId="ocid1\_ipsecconnection.oc1.eu-zurich-1.aaaaaaaalvchpf4vwdngofstjo3xrqnstegj4uv2o5tyw6g2vn5of3oca"]



### Notifications

#### Destinations

Destination service ⓘ  
Notifications Service

Compartment ⓘ  
OCI\_Metrics\_Blog  
git-test (root)/maradule/OCI\_Metrics\_Blog

Topic ⓘ  
OCI\_Metrics\_Blog\_Topic  
[Create a topic](#)

Message Format ⓘ  
☒ Send formatted messages  
☐ Send Pretty JSON messages (raw text with line breaks)  
☐ Send raw messages

[+ Additional destination service](#)

☒ Repeat notification? (if alarm continues to fire) ⓘ  
Notification frequency  

60 minutes

☒ Suppress notifications  
Suppression description ⓘ Optional  

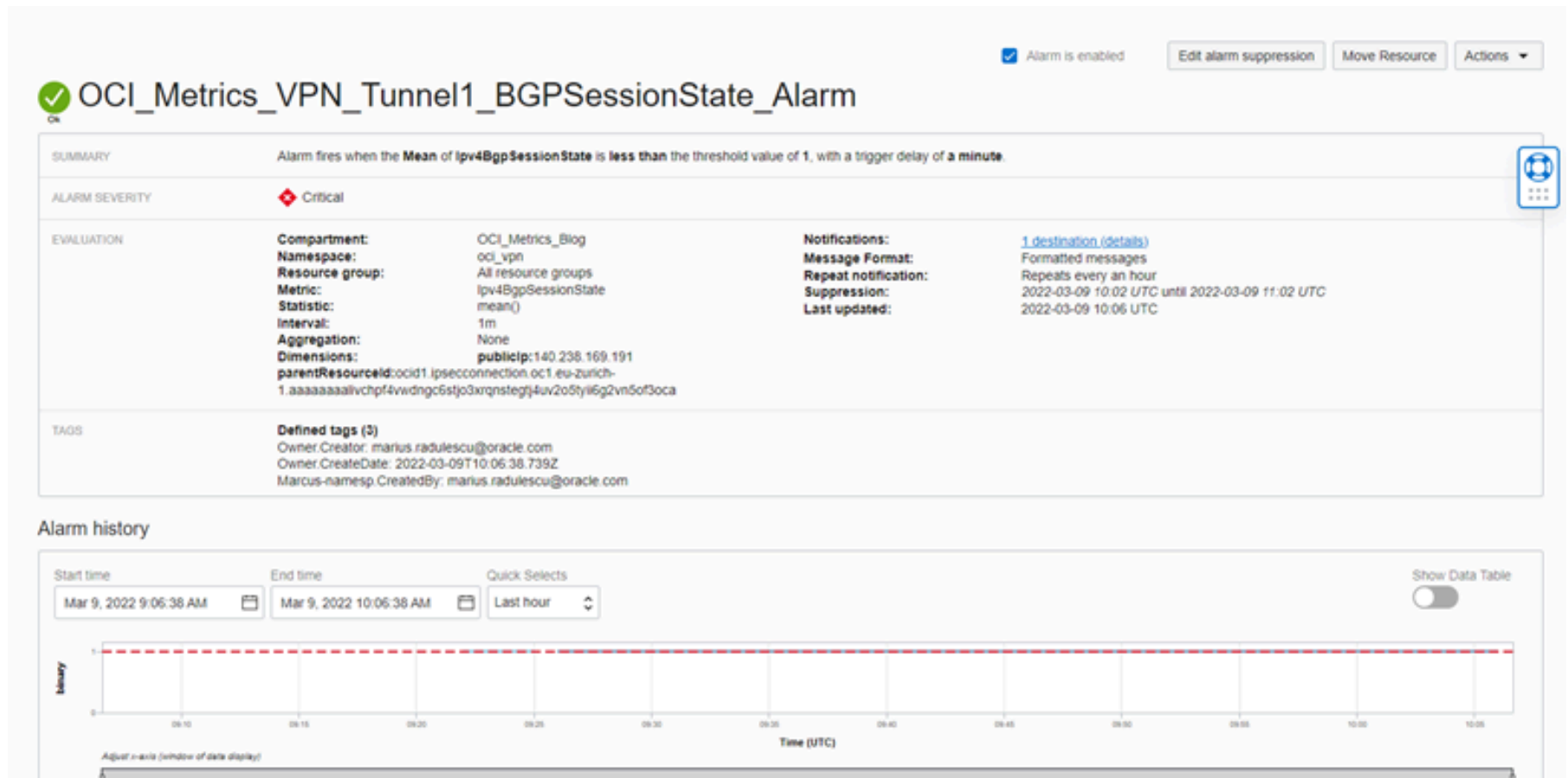
Start time  
Mar 9, 2022 10:02:10 AM

End time  
Mar 9, 2022 11:02:10 AM

☒ Enable this alarm? ⓘ  
This alarm applies to 2 metric streams.  

[Save alarm](#) [Cancel](#)

After creation Alarm page should look like this



Now that we have configured one alarm for FastConnect and one for VPN, we can consider these your templates. We can create multiple alarms with a different metric as a starting point from the metrics list available on the FastConnect or VPN. We need to follow step 2 for Fastconnect or step 3 for VPN to create more alarms.

## Validation

At this point, we should see Alarms firing and sending notifications.

I have chosen to bring the VPN BGP Session down from the CPE device for this validation. Once I have done this, I am seeing see the following:

BGP session on the VPN tunnel goes down

Tunnels in OCI\_Metrics\_Blog Compartment

| Name                             | Lifecycle State ⓘ | IPSec Status ⓘ | VPN BGP Status ⓘ | IPV6 BGP Status ⓘ | Oracle VPN IP Address | Routing Type        |
|----------------------------------|-------------------|----------------|------------------|-------------------|-----------------------|---------------------|
| <a href="#">Tunnel1 - BGP</a>    | Available         | Up             | Down             | Down              | 148.230.169.191       | BGP Dynamic Routing |
| <a href="#">Tunnel2 - Static</a> | Available         | Up             | -                | -                 | 162.67.68.167         | Static Routing      |

Showing 2 Items

Alarm goes on fire status

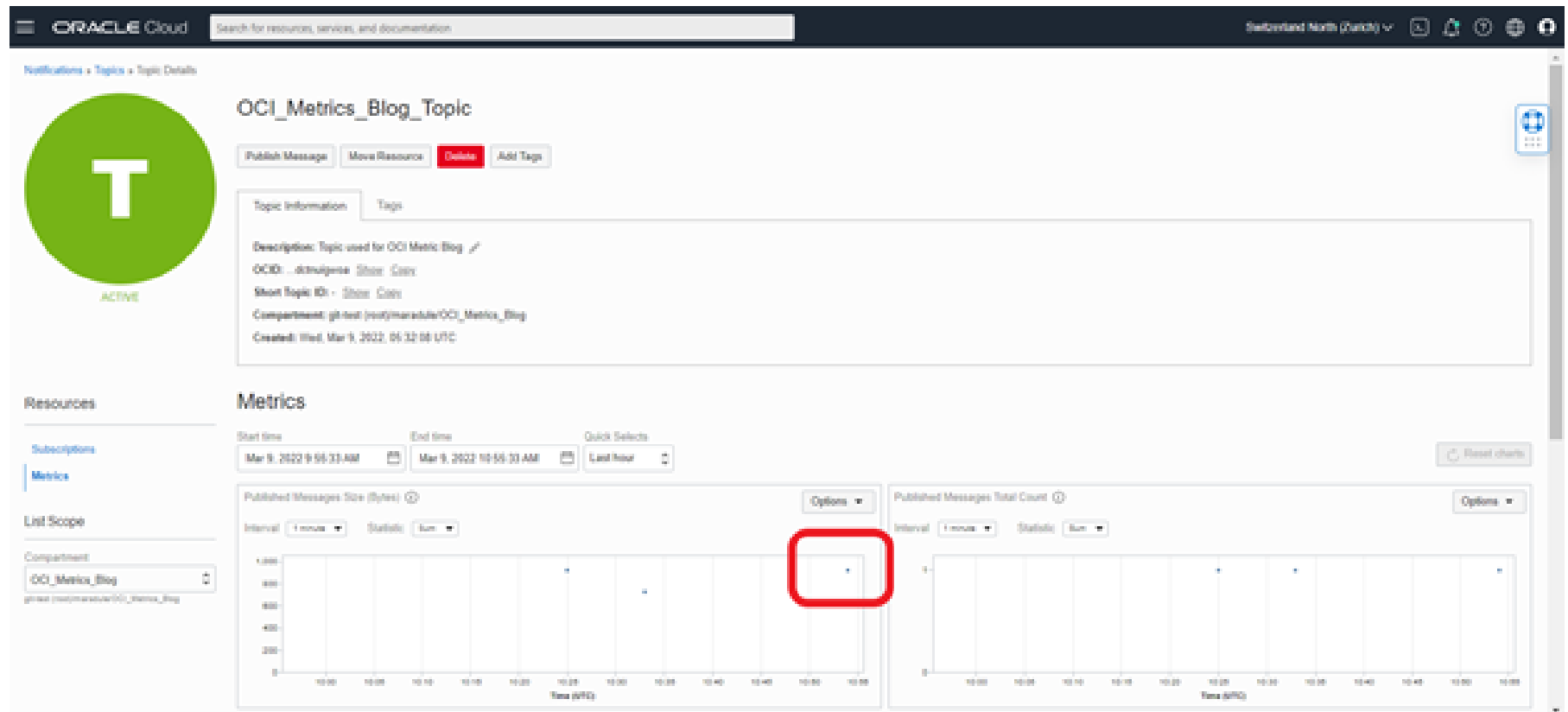


Alarm is triggered

1 alarm is firing in OCI\_Metrics\_Blog Compartment

| OCI_Metrics_VPN_Tunnel1_BGPSessionState_Alarm |                  |                      |            |
|-----------------------------------------------|------------------|----------------------|------------|
| Create Alarm                                  | Add suppressions | Remove suppressions  | Search     |
| Alarm name                                    | Severity         | Triggered time       | Suppressed |
| OCI_Metrics_VPN_Tunnel1_BGPSessionState_Alarm | Critical         | 2022-03-09 10:50 UTC | No         |
| Showing 1 item < 1 of 1 >                     |                  |                      |            |

Notification is sent



An email is received

## OCI\_Metrics\_VPN\_Tunnel1\_BGPSessionState\_Alarm



noreply@notification.eu-zurich-1.oci.oraclecloud.com  
To: Marius Radulescu

Reply Reply All Forward ...

miercuri 09.03.2022 12:53

If there are problems with how this message is displayed, click here to view it in a web browser.



**Alarm Name:**  
OCI\_Metrics\_VPN\_Tunnel1\_BGPSessionState\_Alarm

**Severity**  
CRITICAL

**Status**  
FIRING

**Body**  
null

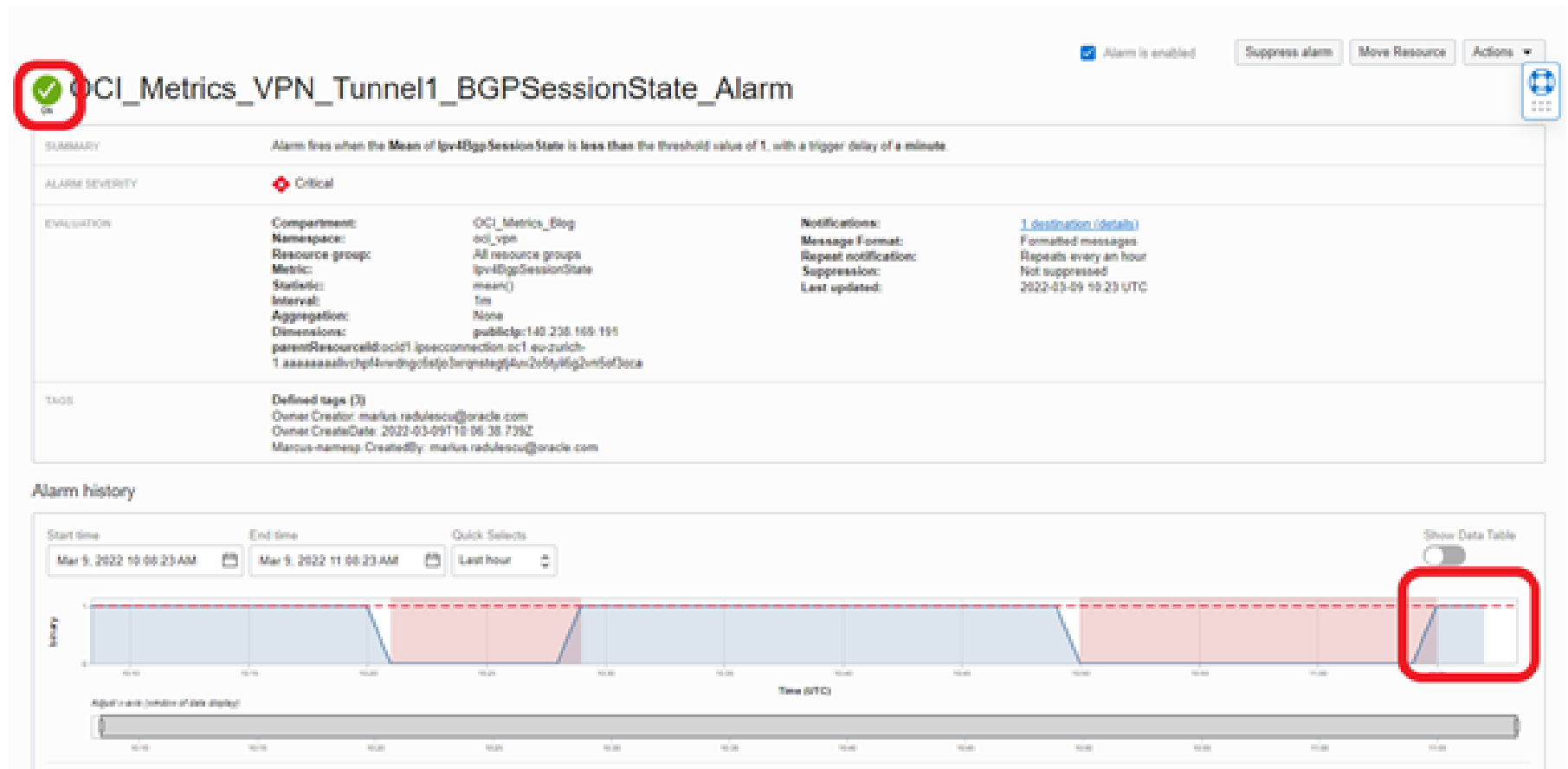
**Time**  
2022-03-09T10:50:00Z

**Query**  
ipsecSessionState[1m][publicIp = "140.230.169.191", parentResourceId = "ocid1\_ipsecconnection\_oci.eu-zurich-1.aaaaaaaalvchpf4vudngc88p3xrgnsteg4uv2o5yil6g2vns0f3oca"].mean() < 1

**Alarm OCID**  
ocid1\_alarm.oci.eu-zurich-1.aaaaaaaakdyke3wjuke2bdtmae6wglouyif7savoomqntpoary2t46d9q

**Type**  
OK\_TO\_FIRING

Once the BGP Session is reestablishing, the Alarm goes on OK status



and a new email is received with the OK status

OCI\_Metrics\_VPN\_Tunnel1\_BGPSessionState\_Alarm

noreply@notification.eu-zurich-1.oci.oraclecloud.com  
To Marius Radulescu

if there are problems with how this message is displayed, click here to view it in a web browser.

ATT00001.htm  
1 KB

ORACLE  
Cloud Infrastructure

Alarm Name:  
OCI\_Metrics\_VPN\_Tunnel1\_BGPSessionState\_Alarm

Severity  
CRITICAL

Status  
OK

Body  
null

Time  
2022-03-09T11:05:00Z

Query  

```

    ipv4BgpSessionState[1m][publicIp = "140.238.169.191", parentResourceId =
    "ocid1.bgpconnection.oc1.eu-zurich-1.aaaaaaahvchp4vrvdrbc9d9c3orgndegj4uv2o5y46g2vn5oD0ca") mean() < 1
  
```

Alarm OCID  
ocid1.alarm.oc1.eu-zurich-1.aaaaaaahvchp4vrvdrbc9d9c3orgndegj4uv2o5y46g2vn5oD0ca

Type  
FIRING\_TO\_OK

Number of metrics breaching threshold  
0

miercuri 09.03.2022 13:08

Reply Reply All Forward

## Caveats and Limitations

Fastconnect is pulling raw data from infrastructure at 4 minutes interval

Fastconnect with a partner "ConnectionState" will go down only if something happens with the partner links to OCI and is not verified per customer.

There is no "bandwidth throughput" on VPN tunnels

In traffic volume, for VPN, is in Bytes, not in Bits as it should be for network traffic



**Marius Radulescu**

Principal Cloud Solution Architect

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